

Erdős Problem #271

EP-271

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $A(n) = \{a_0 < a_1 < \dots\}$ be the sequence defined by $a_0 = 0$ and $a_1 = n$, and for $k \geq 1$ define a_{k+1} as the least positive integer such that there is no three-term arithmetic progression in $\{a_0, \dots, a_{k+1}\}$.

Can the a_k be explicitly determined? How fast do they grow?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>